

Educational Productivity Network (EPN)

Technology Stack Specification & Infrastructure Architecture

Document Type: Architecture & Tech Stack

Status: Technical Draft

Date: June 2026

1. Architectural Constraints & Foundations

The technology stack for the Educational Productivity Network (EPN) is explicitly driven by three core architectural principles defined within the system design requirements: **Local/Edge Deployment** (to fulfill student data-privacy regulations like FERPA), **Multi-Agent Governed Autonomy** (separating execution from cognition), and an **Institution-Controlled Permissioned Ledger** (providing immutable data provenance and tamper-evident audit logs).

Mandatory Compliance Target: Raw behavioral telemetry and Personally Identifiable Information (PII) for minors must remain localized on-premises or within edge nodes inside the school network. Only cryptographic hashes and non-identifiable minimal metadata are permitted to leave the local node environment.

2. Tiered Technology Stack Selection

2.1 Core Multi-Agent & Cognition Layer

This layer powers the core "Perceive-Reason-Plan-Act-Observe-Adapt" cycle through specialized autonomous frameworks, handling orchestrations and localized inference execution.

- Python 3.11+** : Selected as the primary execution runtime for the autonomous agents, leveraging its mature ecosystem for machine learning, data processing, and graph networks.
- LangGraph / CrewAI** : Utilized to structure the hybrid multi-agent system (MAS). LangGraph is chosen specifically to orchestrate cyclical state machine loops required by the continuous 6-stage adaptive learning loop, while managing state handoffs between the 5 distinct agents.
- Ollama / vLLM (Edge Deployment)** : Localized LLM inference engines deployed directly on school edge hardware. Deploys quantized instances of *Llama-3-8B-Instruct* or *Mistral-7B-Instruct* to eliminate cloud dependency, minimize token costs, and maintain zero-data-leakage compliance.
- NetworkX** : Employed by the **Diagnostic Agent** to manage the localized per-student conceptual knowledge graphs and by the **Curriculum Planner Agent** to handle and update dynamic task graphs.

2.2 Local Edge & Data Storage Layer

This layer isolates sensitive behavioral interaction telemetry, local agent session histories, and specific content bodies off-chain within institutional boundaries.

- PostgreSQL + PGVector** : The primary relational database deployed on local school clusters. PostgreSQL manages structured telemetry and student profiles. The PGVector extension powers the

Episodic Memory Store, allowing semantic retrieval of long-term term progress records, past interventions, and student modalities.

- **Redis** : Configured for fast key-value storage to handle the **Working Memory** of running agent sessions, persisting real-time clicks, pauses, and retry streams safely until session cycles resolve.

2.3 Permissioned Distributed Ledger Layer

An append-only institution-controlled consensus network provides the mandatory accountability trail without the overhead or exposure of a public chain.

- **Hyperledger Fabric** : Selected as the enterprise-grade permissioned ledger framework. Its support for private data collections, channels (for district vs. multi-school multi-tenancy), and minimal transaction latency aligns with the Phase 2 and Phase 3 roadmap specifications.
- **Go / TypeScript (Chaincode)** : Used to develop smart contracts establishing automated policy gates for verifiable achievement milestones and credential issuances.

2.4 Content Validation & Interoperability Layer

- **Pydantic Logfire / Guardrails AI** : Implements the central **Policy-as-Code Gateway**. Validates generated text from the Content Generation Agent against deterministic JSON schemas, age-appropriateness filters, and standard curriculum bodies before rendering materials.
- **LTI 1.3 (Learning Tools Interoperability)** : Standardized integration framework utilized to securely ingest performance scores, telemetry streams, and push generated materials directly to existing LMS infrastructure (e.g., Canvas, Moodle).

3. Architectural Mapping of Tech Stack to EPN Agents

The five functional agents map distinct components of the technology stack to preserve the decoupled "cognition vs. execution" design philosophy:

Agent Name	Tech Stack Components	Implementation Responsibility
Monitor Agent (Reactive)	LTI 1.3 Streaming, Redis, FastAPI Telemetry Endpoints	Real-time ingestion of assessment scores and fine-grained engagement telemetry. Triggers circuit-breakers on error spikes.
Diagnostic Agent (Deliberative)	NetworkX (Knowledge Graphs), Local SLM Embeddings	Evaluates student states against conceptual nodes, classifies root-causes of recurring misconceptions.
Curriculum Planner (Planning)	LangGraph ReAct Loops, NetworkX Task Graphs	Decomposes term goals, modifies task sequence, and routes high-impact adjustments to Teacher HITL queue.
Content Generator (Action / Tooling)	vLLM Inference, Guardrails AI, WeasyPrint / Markdown Engine	Produces tailored multi-modal materials. Hashes all outputs instantly and routes them to validation gates.
Orchestrator Agent (Reasoning Engine)	Python State Framework, PostgreSQL, Fabric SDK	Coordinates calls across loop stages, manages budgets, commits verified hashes and actions to the ledger layer.

4. Release Phase Infrastructure Transition

The tech stack expands sequentially across the release phases defined in the Product Requirements Document:

- **Phase 1: Pilot (Single Classroom):** Run localized on an individual server node. Employs SQLite / PostgreSQL + PGVector and local file logging for agent actions. No ledger layer is present; focus is placed entirely on validating the precision of content validation gates and teacher HITL feedback loops.
- **Phase 2: School Rollout:** Scaling up to support up to 2,000 students per edge deployment. Introduces Hyperledger Fabric on-prem nodes managed by IT administrators. Redis is wired for real-time telemetry, and budgeted autonomy caps are programmatically enforced via Orchestrator scripts.
- **Phase 3: District / Multi-School:** Multi-channel Hyperledger Fabric architectures deployed across schools. Parent/Guardian verification interfaces are unlocked via web portals pulling signed cryptographic credentials directly from the blockchain peer nodes, enabling secure, portable episodic data transfers between institutions.